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ENGINEERING
TEXAS A&M UNIVERSITY

Sound Localization Robot, Team 5

Bi-Weekly Update 3

Samuel Lackey, Sanjana Pasumarthi, Lucas Kettler, Joshua Teele

TA: Muhammad Bin Mughees
Sponsor: Dr. Moreira

Project Scope

Figure 1. First responder crawls through smoke and broken glass to rescue elderly man from house fire (Independent TV)

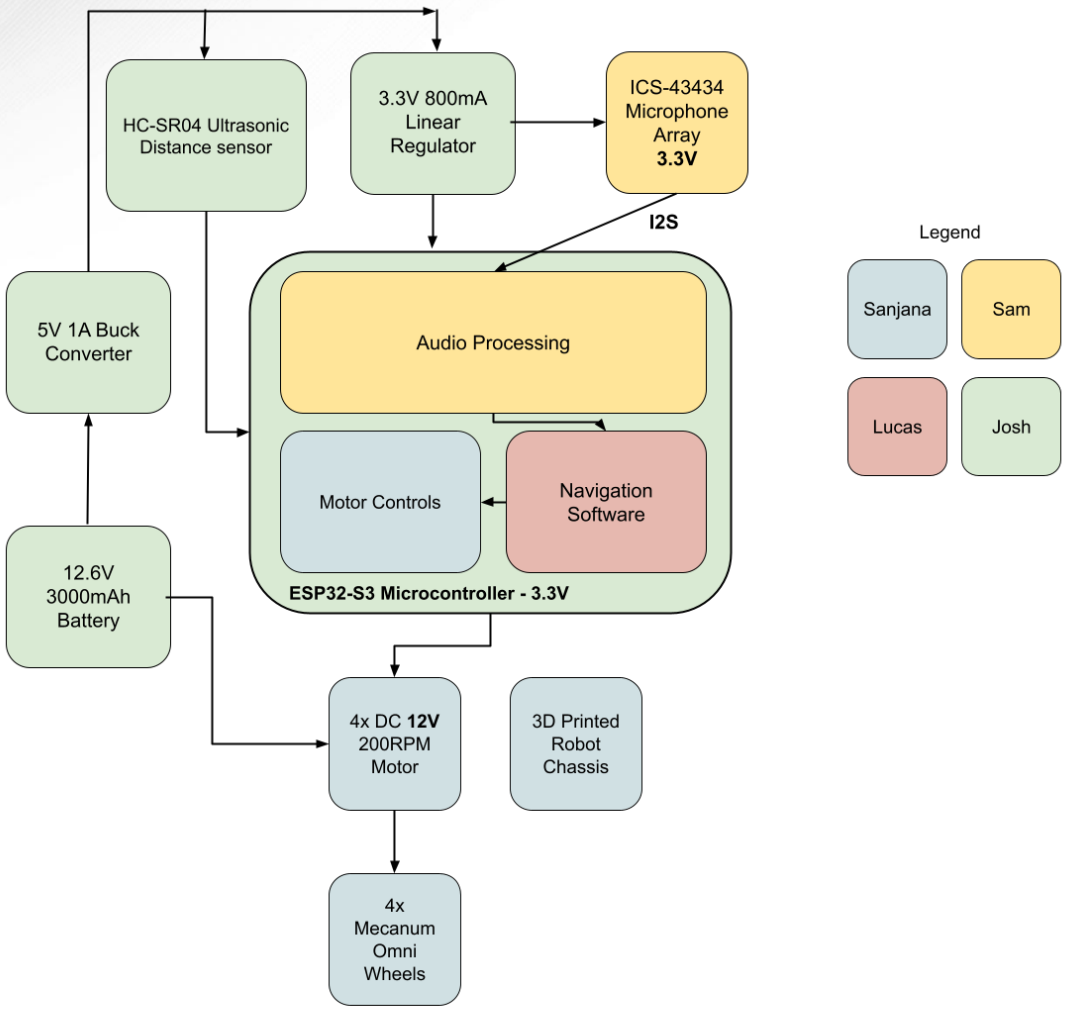
- Keeps **first responders safe** by entering hazardous or toxic areas first.
- **Reduces search and rescue time** by avoiding exploring unnecessary paths
- **Reliable in low-visibility** or high-stress conditions (smoke, debris, noise).

Problem: First Responders lack reliable methods for navigating dangerous environments, where visibility is limited, putting both their safety and the victims at risk.

Our Solution: Building a Sound Localization robot that can hear distress calls, pinpoint their direction, and move toward them autonomously, allowing first responders to focus on saving lives instead of risking theirs.

*A smarter, safer, and faster way to locate victims, all while protecting rescuers and saving more lives.

Robot Sub Systems:



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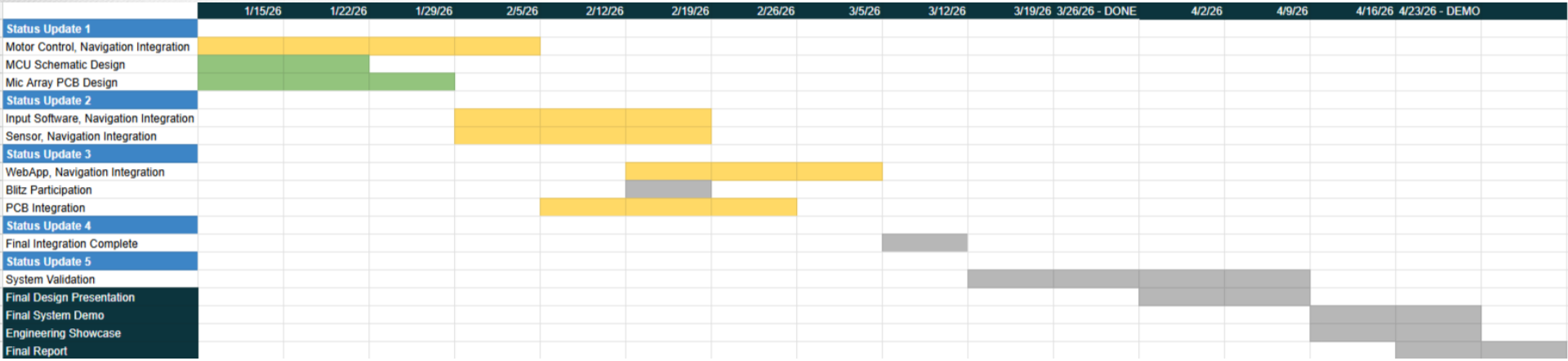
Project Timeline



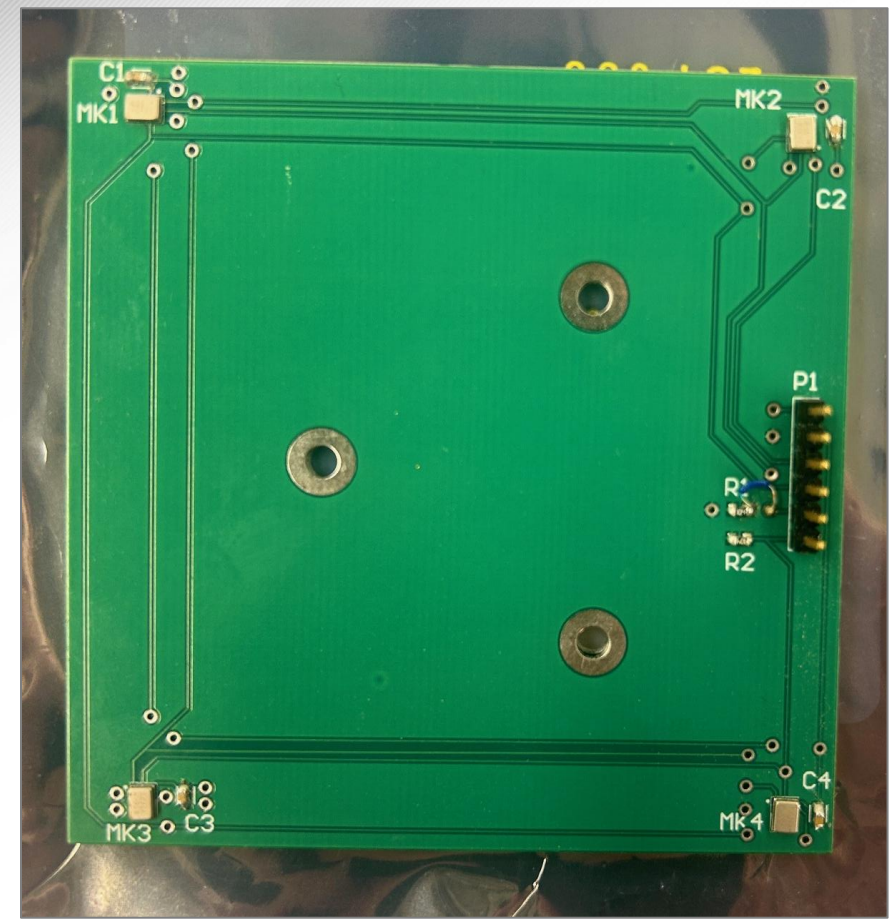
Input Signal Software

Samuel Lackey

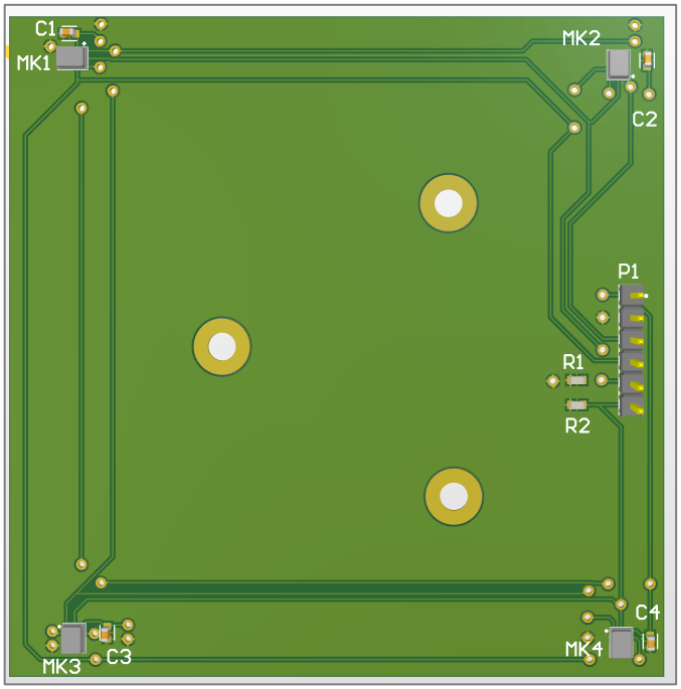
Accomplishments since last update : 15 Hours of Effort	Ongoing progress/problems and plans until the next presentation
• Completed Microphone Array PCB • Adapted Sound Localization Algorithm into a single callable function	• Fully incorporate Signal Processing code into Navigation subsystem • The new Microphone PCB is not fully functional yet

Input Signal Software

Samuel Lackey



Microphone Array PCB



Microphone Array PCB 3D Design

Only 2/4 Mics are outputting data
- Check solder connections
- Resolder non-functional mics

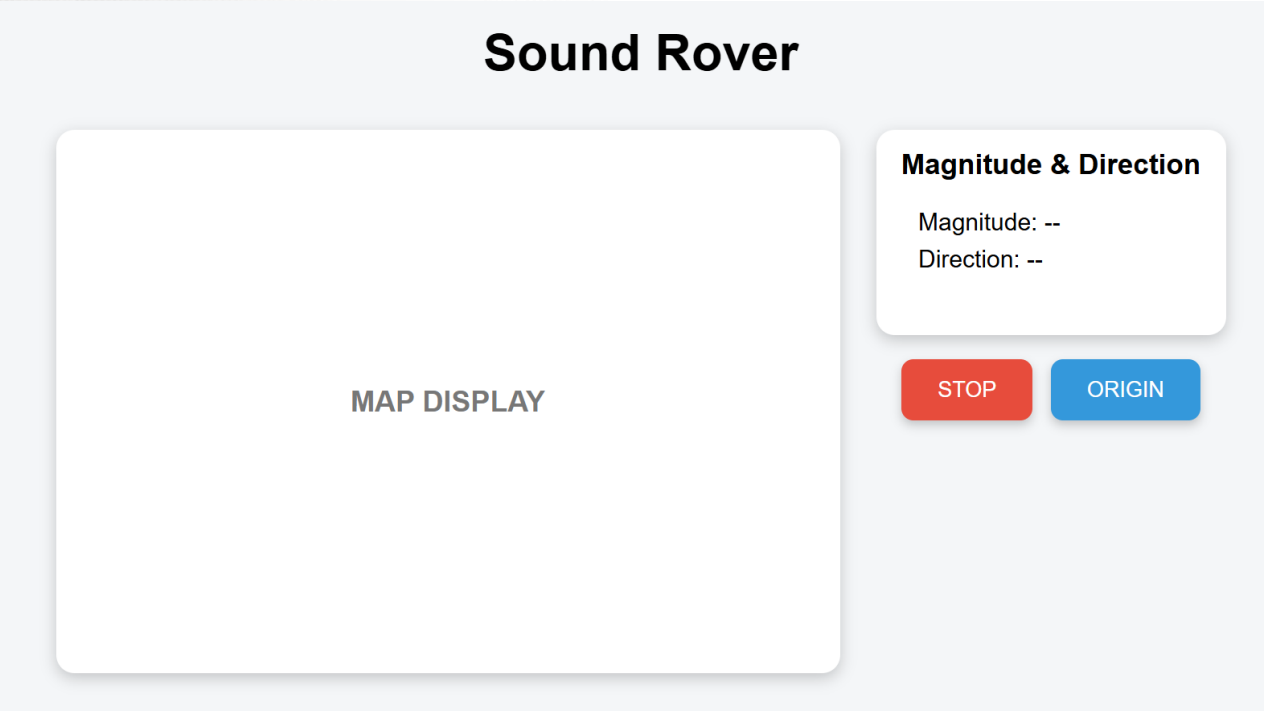
Mechanical/ Motor Controls Subsystem

Sanjana Pasumarthi

Accomplishments since Last Update 11 hours of effort	Ongoing progress/problems and plans until the next presentation
• Set back with some of the components failing, had to order replacements and rewire for testing • Basic UI is complete to display information	• Program a method to record pathway trace for the rover to return to its origin • Integrating the controls with Lucas's Code

Mechanical/ Motor Controls Subsystem

Sanjana Pasumarthi



- Final UI design of website accessible through connect of ESP wifi module
- Includes emergency stop, pathway of rover, return to origin button, magnitude and direction of sound source

Navigation Software Subsystem

Lucas Kettler

Accomplishments since Last Update 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Partial integration of Input software/webapp • Integration of Sensor code	• Validate integration of subsystems • Debug integration components

Navigation Software Subsystem
Lucas Kettler

Power & Distance Sensor Subsystem

Joshua Teele

Accomplishments since Last Update 10 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Solder PCB and test with multimeter • Tested distance sensor code on new MCU	• Test rest of individual subsystems on new MCU • Ongoing: Integrate distance sensor code with navigation software

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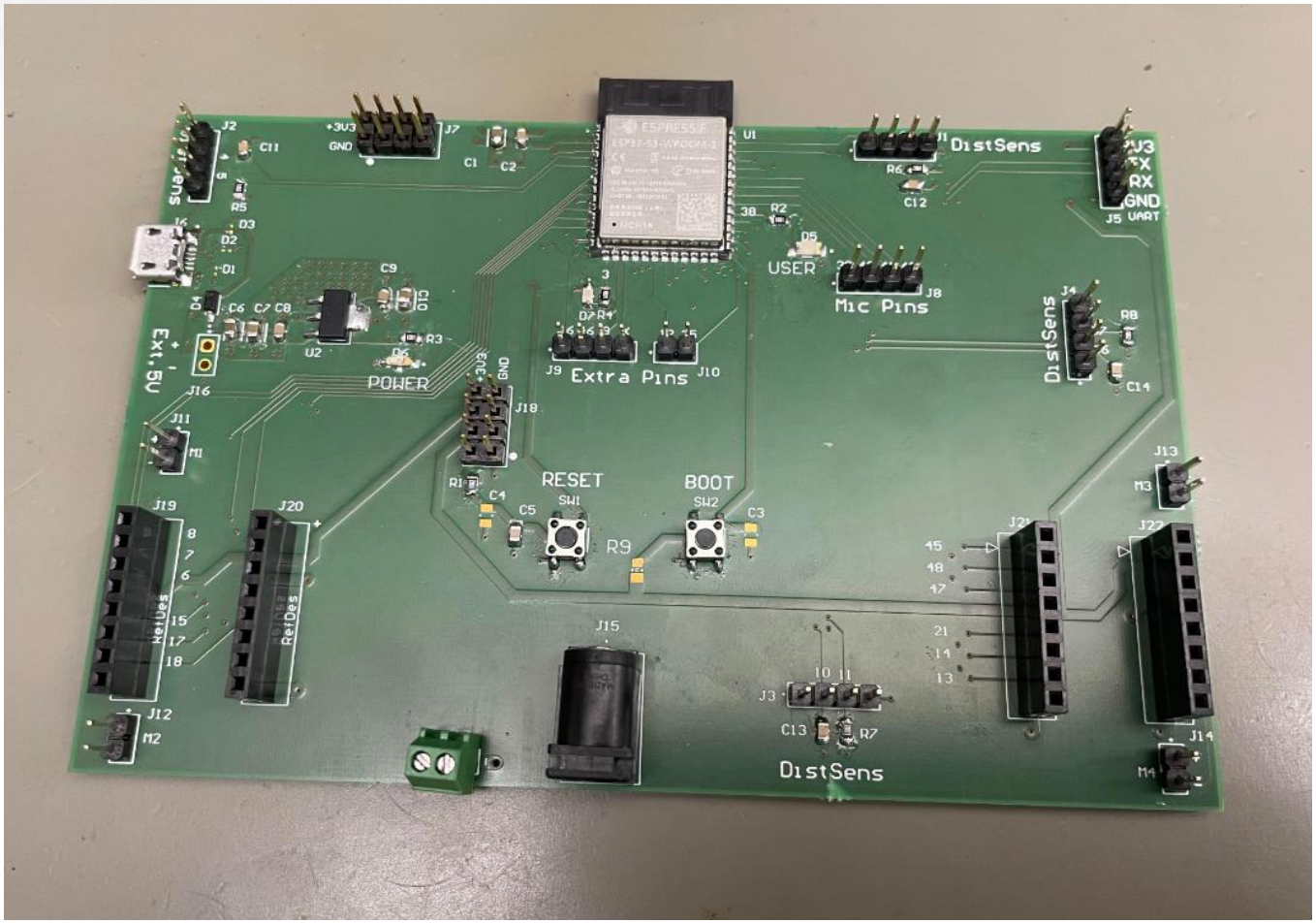
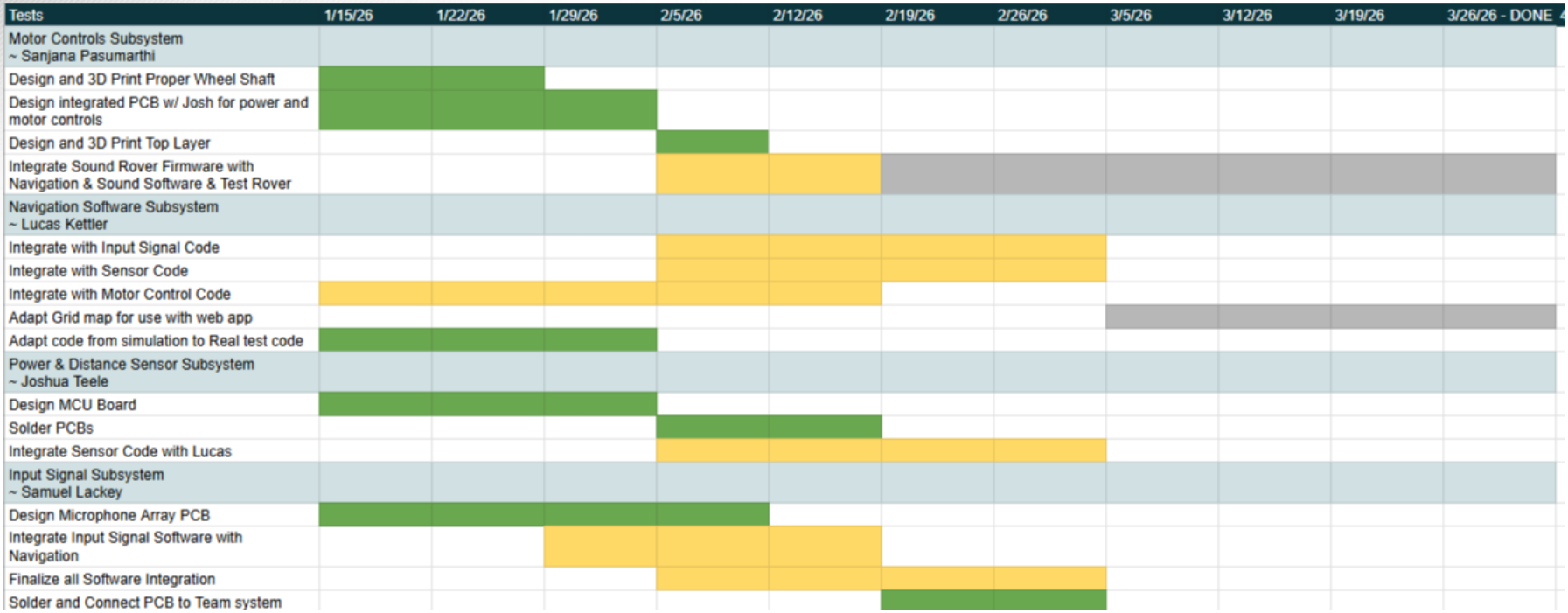
Power & Distance Sensor Subsystem

Joshua Teele



Execution Plan



Validation Plan

Paragraph	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.1	DOA Angle Accuracy	All 5 tests must log a detected angle within an 3.6°	Place speaker playing 450Hz tone at a known angle, and 0.5m distance. Run the detection test 5 times	UNTESTED	Sam
3.2.1.2	DOA With Background Noise	All 5 tests must log a detected angle within an 3.6°	Place speaker playing 450Hz tone at a known angle, and 0.5m distance while separate 550Hz tone is playing. Repeat 5 times	UNTESTED	Sam
3.2.1.3	Obstacle Detection	Verify robot halts movement at the node prior to an object	Isolated test with robot attempting to drive to a goal behind an obstacle	UNTESTED	Lucas, Joshua, Sanjana
3.2.1.4	Grid Mapping	Verify the robot updates a grid map for the web app	View output of webapp upon completion of path	UNTESTED	Lucas, Sanjana
3.2.1.5	Webapp	The webapp must show the angle, estimated distance, and control the robot.	Run the program and connect to wifi, trigger a sound, access the webapp, and move.	UNTESTED	Sam, Sanjana
3.2.1.6	Distance Estimation	The estimated distance must be less than 0.5m of the true distance	Run the distance calibration software, move sound source from 0.5 to 1m to 2m.	UNTESTED	Sam
3.2.2.1	Accurate Robot Movement	Robot moves in intended direction determined by program	Series of coordination tests for the robot to move to the sound source	UNTESTED	Full Team
3.2.2.2	Constant Response rate of robot	Robot has a constant rate of response with +/- 5% of average time	Multiple trials of measuring time it takes for rover to change status	UNTESTED	Full Team
3.2.2.3	Structurally strong Robot	Wheel shaft does not break easily, no unintended movement from motors	Test robot with obstacles and quick movement	UNTESTED	Sanjana
3.2.3.3.1	Input Voltage(USB)	Ensure voltage at USB port is above 4.75V (minimum needed voltage for linear regulator)	Use multimeter to test output of buck converter	UNTESTED	Joshua
3.2.3.3.2	Input Voltage(DC jack)	Ensure voltage at DC jack is between 10.8 and 13.2 V (+/-10% of motor voltage)	Use multimeter to test output of buck converter	UNTESTED	Joshua
N/A	Full System Demo	Robot is able to navigate autonomously around obstacles, reach goal, and display data on app	Multiple observations and trials of robot	UNTESTED	Full Team

Thank You,
Any Questions?

Sound Localization Robot 5 – Team 23

Lucas Kettler, Samuel Lackey, Joshua Teele,
Sanjana Pasumarthi